

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Industrial and Systems Engineering

2. **Subject Name:** Simulation Lab

L-T-P: 0-0-3

Credits: 2

3. **Pre-requisites:** IM21201: Operations Research - I

4. **Syllabus and reference books:**

Syllabus:

Random number generation using simulation packages, Probability distribution fitting, Testing of random numbers: Chi - square test, Kolmogorov-Smirnov test, Hypothesis testing. Development and testing of simulation models for IE functional systems, Use of discrete and continuous simulation packages. Discrete Event Simulation, Simulation of single server queue and two server queue, Simulation of Inventory system. Continuous Simulation, Development of causal loop diagrams, Development of stock and flow diagrams, Policy experimentations. Validation of simulation models, Extreme condition test, Behaviour validity test: Reference mode behaviour, Behaviour replication test, and Error analysis. Development and experiment with simulation games.

Reference Books:

- 1) Jerry Banks, John S. Carson, Barry L. Nelson, David M. Nicol, Discrete-Event System Simulation, Pearson Education, 2010.
- 2) Averill M. Law, David Kelton, Simulation Modeling & Analysis, McGraw-Hill, 1991.
- 3) John Sterman, Business Dynamics: Systems Thinking and Modeling for a Complex World, McGraw-Hill, 2001.
- 4) Geoffrey Gordon, System Simulation, 2/E, PHI Learning, 2002.

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Random number generation using simulation packages, Probability distribution fitting, Testing of random numbers: Chi - square test, Kolmogorov-Smirnov test, Hypothesis testing.	6
2.	Development and testing of simulation models for IE functional systems, Use of discrete and continuous simulation packages, Discrete Event Simulation, Simulation of single server queue and two server queue, Simulation of Inventory system.	9
3.	Continuous Simulation, Development of causal	

	loop diagrams, Development of stock and flow diagrams, Policy experimentations.	9
4.	Validation of simulation models, Extreme condition test, Behaviour validity test: Reference mode behaviour, Behaviour replication test, and Error analysis.	6
5.	Development and experiment with simulation games.	6
Total number of hours		36